

LEARN PYTHON PROGRAMMING – BEGINNER TO MASTER

Section: Sec 1. Introduction To Python

Lecture: 6. Python is Platform Independent

Section 1: Lecture Summary

The lecture explains the concept of platform independence, particularly how Python is platform independent and portable. It first defines what a platform is (an operating system like Windows, macOS, Linux) and how compiled languages like C produce machine code that is platform-dependent, meaning compiled files only run on the platform they were built on. It then contrasts this with interpreted languages like JavaScript, which run source code in a platform-independent way via interpreters available on all platforms (e.g., browsers). The lecture introduces Python's hybrid nature: Python source code is compiled to platform-independent bytecode (.pyc), which runs on a platform-specific Python Virtual Machine (PVM). This two-step compilation and interpretation process makes Python programs platform independent if the platform has the appropriate PVM installed. The lecture concludes emphasizing the importance of understanding compilers, interpreters, platform independence, and hybrid compilation in programming.

Section 2: Key Concepts and Explanations

- Platform and Platform Independence

- Platform: The operating system environment where applications run (Windows, macOS, Linux).
- Programs compiled for one platform generate machine code specific to that platform and cannot run on others without recompilation.
- Platform independence means the same program can run on multiple operating systems without modification.

- Compiled Languages (e.g., C)

- Source code is translated into machine code (executable) specific to an OS/platform.

- Compiled executables (.exe on Windows) do not run on different platforms.
- To run on another platform, the source code must be recompiled on that platform.

- Interpreted Languages (e.g., JavaScript)

- Source code runs directly on an interpreter present on each platform.
- For JavaScript, the browser acts as the interpreter runtime environment, available on every major platform.
- Pros: Platform independence of source code.
- Cons: Source code visibility to users.

- Python's Hybrid Compilation and Interpretation

- Python source code is compiled once to an intermediate bytecode (.pyc).
- Bytecode is platform independent and cannot be easily read like source code.
- Bytecode runs on a platform-specific Python Virtual Machine (PVM).
- Each platform has its own PVM but provides the same runtime environment.
- Python combines compilation and interpretation steps, making it portable across platforms.

- Runtime Environment

- Like browsers for JavaScript, PVM acts as a runtime environment that interprets Python bytecode.
- Ensures that Python programs run on any OS with a compatible PVM installed.

- Summary of Concepts

- Compiler: Translates source code to machine code.
- Interpreter: Executes source code directly.
- Hybrid: Uses an intermediate bytecode and runtime environment.

- Portability: Ability of a program to run on multiple platforms without modification.

Section 3: Example Code and Use Cases

The lecture is conceptual and does not provide explicit code examples. Below is a reconstruction illustrating the discussed concepts in Python:

Python source code (platform-independent)

When this is run, Python compiles this to bytecode (.pyc)

and executes it in the Python Virtual Machine (PVM),

which is platform-specific but runs the same bytecode.

```
print("Hello, Platform Independent Python!")
```

Explanation:

- The source code is the same on Windows, macOS, Linux.
- Python compiles this to bytecode which runs in PVM optimized for each OS.
- No manual recompilation is necessary like in C.

A conceptual comparison with C:

include <stdio.h>

```
// C source code example (platform-dependent after compilation)
```

```
int main() {  
    printf("Hello, World!\n");  
    return 0;  
}
```

```
// After compiling on Windows, the executable runs only on Windows.
```

```
// To run on macOS or Linux, source must be recompiled on those platforms.
```

Section 4: Key Takeaways

- A **platform** is the OS environment where programs run (Windows, macOS, Linux).
- Compiled languages generate platform-dependent machine code that does not run on other OS without recompilation.
- Interpreted languages run source code on interpreters present on all platforms, offering platform independence but exposing source code.
- Python uses a **hybrid approach**: source code → bytecode (platform independent) → executed by platform-specific Python Virtual Machine (PVM).
- The PVM acts as a runtime environment, enabling Python programs to run unchanged on any OS with Python installed.
- Understanding compilers, interpreters, platform dependence, and hybrid models is fundamental to grasping Python's portability and execution model.

Lecture in Slides

Please Note: This document presents a curated selection of slides from the lecture; others have been excluded for conciseness.





 **Python**
is



 **Python**
is
Platform Independent



Python
is
Platform Independent
Portable

Platform Independent





Platform Independent


Windows


macOS



Platform Independent


Windows


macOS


Linux





























